

3. [Maximum mark: 6]

The loudness of a sound, L , measured in decibels, is related to its intensity, I units, by $L = 10 \log_{10}(I \times 10^{12})$.

Consider two sounds, S_1 and S_2 .

S_1 has an intensity of 10^{-6} units and a loudness of 60 decibels.

S_2 has an intensity that is twice that of S_1 .

(a) State the intensity of S_2 . [1]

(b) Determine the loudness of S_2 . [2]

The maximum loudness of thunder in a thunderstorm was measured to be 115 decibels.

(c) Find the corresponding intensity, I , of the thunder. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

